

Document 1 — System Architecture & How It Works

The complete LangGraph self-hosted AI platform: what we use, how the pieces fit, and the end-to-end flow.

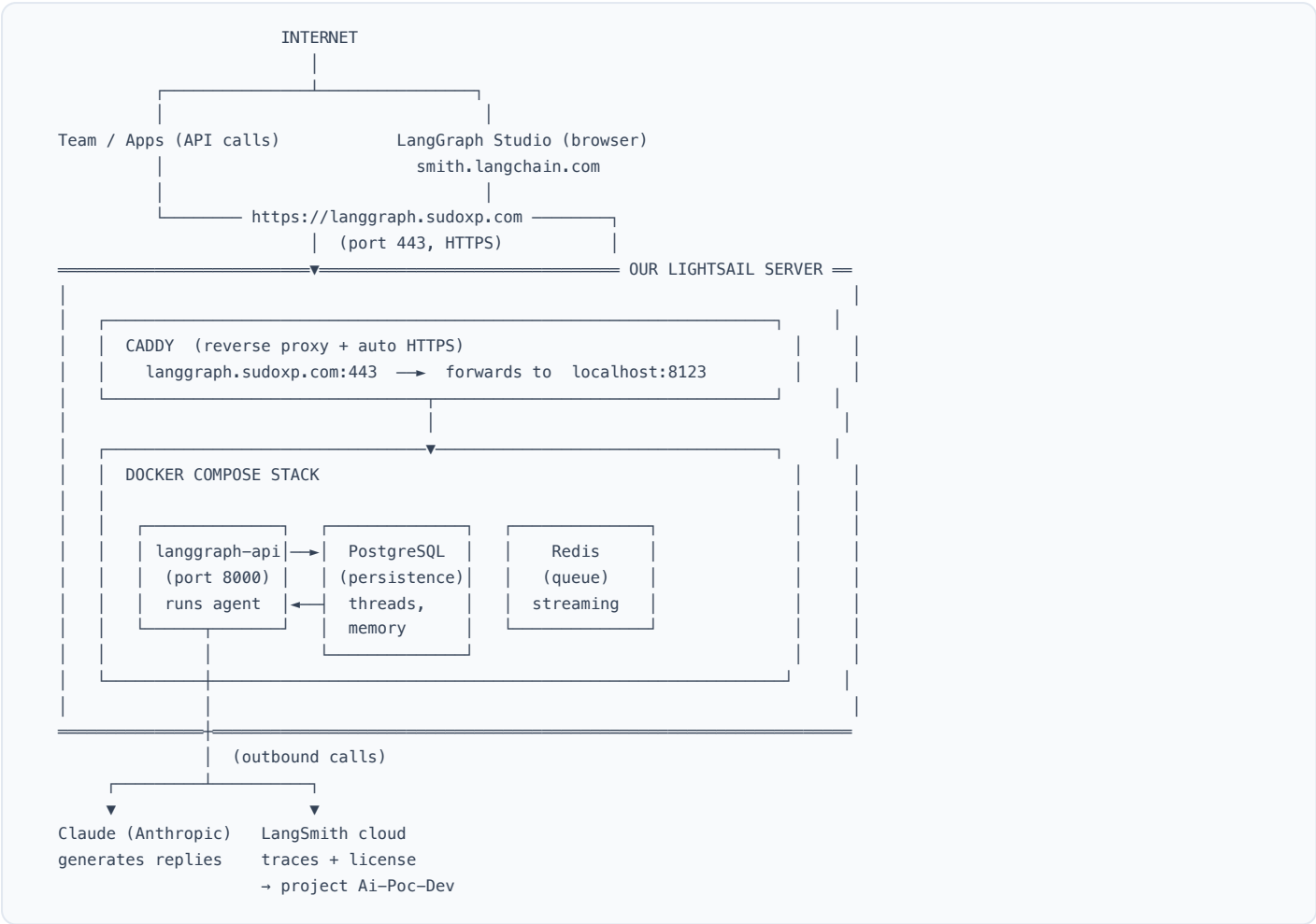
Project: AI-Poc-Dev · Domain: langgraph.sudoxp.com · Repo: github.com/Harikarthik01/LangGraph-Development · Model: claude-sonnet-4-6 · Updated 2026-07-07

This document explains **every technology in our AI platform**, what each does, and **how a request flows** through the system from start to finish. It reflects the **live production setup**: a self-hosted LangGraph server with persistence (Postgres), a task queue (Redis), and HTTPS (Caddy) — all running on our own AWS Lightsail server.

§0 Contents

PostgreSQL	A database that permanently saves each conversation's state. This is what lets agents remember across restarts and gives each user a separate "thread."
Redis	Fast in-memory store used for the task queue (handling many requests), streaming responses, and background jobs.
Caddy	Sits in front of the LangGraph server. It provides HTTPS (secure encrypted access) with an automatic free certificate, and forwards secure traffic to the app. This is what makes <code>https://langgraph.sudoxp.com</code> work.
Docker	Runs LangGraph + Postgres + Redis as three neat "containers" that start together with one command (<code>docker compose up</code>).
LangSmith	Two jobs: (1) licenses our self-hosted server for free, (2) records every run so we can debug it. Traces appear under project Ai-Poc-Dev .
LangGraph Studio	The visual dashboard to build/test agents. Hosted by LangChain; connects to our server over HTTPS.

§3 The complete architecture



Conversation state / memory / threads	PostgreSQL on our server	✓ 100% ours
HTTPS certificate	Our server (Caddy)	✓ ours (free)
Traces (debug logs)	LangSmith cloud	⚠ copy sent (optional)
Studio UI	LangChain cloud	⚠ borrowed, free
AI replies (thinking)	Anthropic (Claude)	✗ needed for replies

§6 Access points & environments

What	URL / Address	Use
Secure API (HTTPS)	https://langgraph.sudoxp.com	Apps, team, Studio ⭐
API docs (Swagger)	https://langgraph.sudoxp.com/docs	Explore/test endpoints
Health check	https://langgraph.sudoxp.com/ok	Is it alive?